

A Semantic Model for Reusable Questionnaires and Temporal Answer Validity in Healthcare

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Abstract

Validated clinical questionnaires remain predominantly distributed as document-based artifacts (e.g., PDFs, electronic forms, or paper questionnaires), treating each instrument as an isolated resource even when the same validated question appears across multiple instruments. This document-centric representation, combined with the lack of a formal model for answer validity over time, forces patients to repeatedly answer the same questions at every appointment, directly contributing to survey fatigue and reducing data quality. This paper presents the FAQIR Questionnaire Ontology (QO), a semantic model that transforms questionnaire documents into knowledge graphs improving FAIRness and where questions become independent, globally identifiable, and reusable entities. Explicit temporal validity metadata (e.g., "valid for 6 months" or "valid until smoking cessation") is attached to questions and their contexts, enabling a reasoning engine to determine whether a previously given answer remains trustworthy at the time of a new questionnaire. When combined with patient consent expressed via the Open Digital Rights Language and the Data Privacy Vocabulary (ODRL-DPV), questions with still-valid answers can be automatically deferred, thereby shortening questionnaires and reducing patient burden. The model is demonstrated through a proof-of-concept implementation using the WHOQOL-BREF instrument and synthetic patient data.

Keywords

FAIR principles, Questionnaire reuse, Temporal validity, Survey fatigue, Person-centric healthcare, Real-World Data

1. Introduction

Patient-reported outcomes remain the gold standard for capturing subjective health states, yet widespread reliance on paper-based and document-centric digital questionnaires has led to user exhaustion and response fatigue [1, 2]. In healthcare, patients face a repetitive burden: the exact same questions are asked at every appointment, contributing to question fatigue, a subtype of survey fatigue that leads to high drop-out rates and low-quality data [2, 3]. Clinical workflows that prioritise repetitive patient intake screening to meet performance metrics often have significant unintended consequences, distracting from high-value care while yielding substandard data quality [3].

Two structural limitations drive this ongoing burden: the lack of a formal model for **temporal answer validity**, and the absence of machine-readable, reusable question semantics. Current practice leaves temporal validity implicit: when a patient reports "I have no problems doing my usual activities", systems have no mechanism to determine how long that statement remains trustworthy. Consequently,

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questions are re-administered at subsequent visits simply because "it is time to ask again," rather than because a model has determined the prior answer expired. Furthermore, digital transformation efforts have largely focused on digitizing paper questionnaires into web forms, PDFs, or isolated electronic data capture templates (e.g., in REDCap)[4, 5]. Traditional workflows distribute translations as separate language-specific PDFs or isolated web forms, creating significant versioning drift and complicating cross-linguistic studies. Modeling questionnaires as Knowledge Graphs leverages native Resource Descriptive Framework (RDF[6]) semantics and language tags (e.g., @en, @es), allowing all localized labels and instructions to reside within a single, unified entity without breaking underlying concept identifiers. Even when validated instruments (e.g., EQ-5D [7], WHOQOL-BREF [8]) are deposited in repositories like the National Institutes of Health Common Data Elements (NIH CDEs) or the Grid-Enabled Measures platform [9, 10], they are catalogued by psychometric constructs without exposing machine-readable, question-level semantics. As a result, when the same validated question (e.g., "Do you smoke?") appears across cardiovascular, respiratory, and pre-operative intake forms, each system treats it as a new, isolated, single-use entry rather than as a reusable component.

To situate FAQIR's contribution, existing representational approaches have been reviewed. The current state-of-the-art for *exchanging* healthcare data is HL7 FHIR (Health Level Seven International Fast Healthcare Interoperability Resources) [11]. While FHIR defines a robust set of resources (*Questionnaire*, *QuestionnaireResponse*) and supports terminology bindings, it is primarily designed as a client-server exchange format rather than a longitudinal graph model (where data is collected sequentially over days, months or years for the same individual). When expressed in RDF, FHIR relies heavily on blank nodes for core structural components, which hinders data discoverability, prevents global identification of individual answers, and complicates temporal reasoning over patient histories [12]. FAQIR eliminates blank nodes across all primary domain entities (questionnaires, sections, questions, wrappers, and answers), permitting anonymous nodes strictly for inline data-type parameter blocks (e.g., `fhir:Coding`).

Beyond FHIR, ontologies like QUESTO [13] and QuestionOnto [14] have explored structural models for survey components. However, both conflate self-reported survey replies with objective clinical observations within their class hierarchies, and QuestionOnto embeds contextual metadata (e.g., positional layout) directly within answers rather than using decoupled wrappers. Crucially, neither framework provides a model for managing answer validity over a patient's longitudinal health journey, nor do they support automated reasoning to defer redundant questions (a detailed architectural comparison is provided in Section 5.1).

This paper addresses these gaps by introducing the **FAQIR Questionnaire Model**: a lightweight ontology and rule framework that transforms document-centric survey instruments into FAIR Knowledge Graphs (KGs). FAQIR shifts the paradigm from single-use questionnaire administration toward integrated, patient-centric answer management. Specifically, this paper makes the following key contributions:

- **Decoupled Semantic Architecture:** It provides a machine-readable representation that decouples atomic, context-agnostic `Question` entities from contextual `OrderedQuestion` wrappers, enabling global question identification and cross-instrument reuse.
- **Explicit Temporal Validity Model:** It introduces machine-actionable validity metadata attached to question wrappers (e.g., valid for 1 day, or until an intervening health event occurs), establishing a formal basis for longitudinal answer trustworthiness.
- **Intelligent Question Deferral:** It enables an automated reasoning service that evaluates a patient's existing valid answers against new survey requirements, automatically deferring redundant questions to mitigate survey fatigue.
- **FAIR Implementation and Validation:** It delivers an automated, reproducible build pipeline generating open-source OWL[15] and SHACL[16] artifacts,¹ validated through a full-scale semantic implementation of the WHOQOL-BREF instrument evaluated over synthetic patient data.

¹<https://github.com/faqirfoundation/questionnaire-ontology>

The remainder of this paper is organized as follows: Section 2 outlines the ontology design requirements and competency questions; Section 3 details the FAQIR Questionnaire Ontology architecture and LinkML[17] implementation; Section 4 presents the WHOQOL-BREF validation use case and SPARQL Protocol and RDF Query Language[18] query execution; Section 5 compares FAQIR against related work; and Section 6 concludes with future directions.

2. Ontology Design Requirements and Competency Questions

The engineering of the FAQIR Questionnaire Ontology (QO) followed a formal, goal-oriented requirements analysis grounded in the LOT methodology [19]. To address the structural and operational gaps in existing survey formats, the ontology was designed around five core functional requirements (FR1–FR5):

- **FR1 (Linked Data & Semantic Web Grounding):** The model must adhere strictly to Resource Description Framework (RDF) standards, employing global Unique Resource Identifiers (URIs) for all primary entities and answers to prevent graph dislocation and enable point-in-time SPARQL traversal.
- **FR2 (Atomic Question Decoupling):** Questions must be modeled as context-agnostic, globally reusable atomic entities (`qo:Question`) distinct from their survey-specific layouts.
- **FR3 (Atomic Section Grouping):** Structural survey partitions must be represented as reusable atomic sections (`qo:Section`) to enable modular instrument composition.
- **FR4 (Diverse Input Support):** Inquiry items must accommodate multiple input paradigms, including single/multiple choice, open choice, bounded numeric intervals, decimals, and free text.
- **FR5 (Machine-Actionable Temporal Validity):** The model must specify contextual lifespan boundaries (`time:Duration`) and enforcement modes (`hardValidity`) for answer trustworthiness.

The functional requirements were operationalized into formal Competency Questions (CQs) used to verify schema expressiveness and guide SPARQL query execution:

- **CQ1 (Composition):** Which atomic questions and sections constitute a given questionnaire template, and what are their positional sequence indices?
- **CQ2 (Datatype Constraints):** What input datatype, permissible coding options, or numeric range parameters govern a specific atomic question?
- **CQ3 (Temporal Validity):** Is a patient’s previously recorded answer to an atomic question still valid at the timestamp of a new survey session?
- **CQ4 (Deferral Eligibility):** Which questions in a new questionnaire can be automatically deferred or pre-filled based on valid, consent-authorized historical answers?
- **CQ5 (Cross-Instrument Analytics):** Which distinct validated instruments share the same atomic question or SNOMED CT[20] clinical concept tag?

3. FAQIR Questionnaire Ontology Architecture

The FAQIR Questionnaire Ontology (QO) is designed to support patient-centric management of questionnaires and their responses, enabling the tracking of a patient’s longitudinal health journey. While the FAIR principles (Findable, Accessible, Interoperable, Reusable) provide the foundational framework, FAQIR places special emphasis on *Interoperability* and *Reusability* with *Quality* as the central guiding principle (FAIR+Q).

A key distinction in QO is the separation of *objective measurements* (e.g., weight, blood pressure) and *personal data* (e.g., date of birth), which are better captured as observations or demographic attributes, from *questionnaires* (e.g., EQ-5D [7] and WHOQOL-BREF [8]). While “everything could be a question” in principle, FAQIR focuses on structured, validated instruments where the question-answer pattern is clinically meaningful and where temporal validity reasoning can actively reduce patient survey fatigue.

3.1. Development Methodology and Open Artifacts

The QO was developed following an iterative, expert-informed engineering process grounded in established ontology development methodologies, such as LOT [19] and SAMOD [21].

To ensure complete transparency, reproducibility, and FAIR compliance, all development artifacts are hosted in an open-source GitHub repository². The repository contains the canonical LinkML source schema, compiled OWL (.owl.ttl) and SHACL (.shacl.ttl) files, Markdown documentation with embedded Mermaid architecture diagrams, instance graphs for validated instruments, and executable SPARQL competency queries.

3.2. Core Classes and Conceptual Hierarchy

All original FAQIR terms reside in the stable namespace <https://ns.fair.org/q-o#> (prefix qo:), maintained by the FAQIR Foundation. The structural architecture and upper-ontology grounding of QO are illustrated in Figure 1.

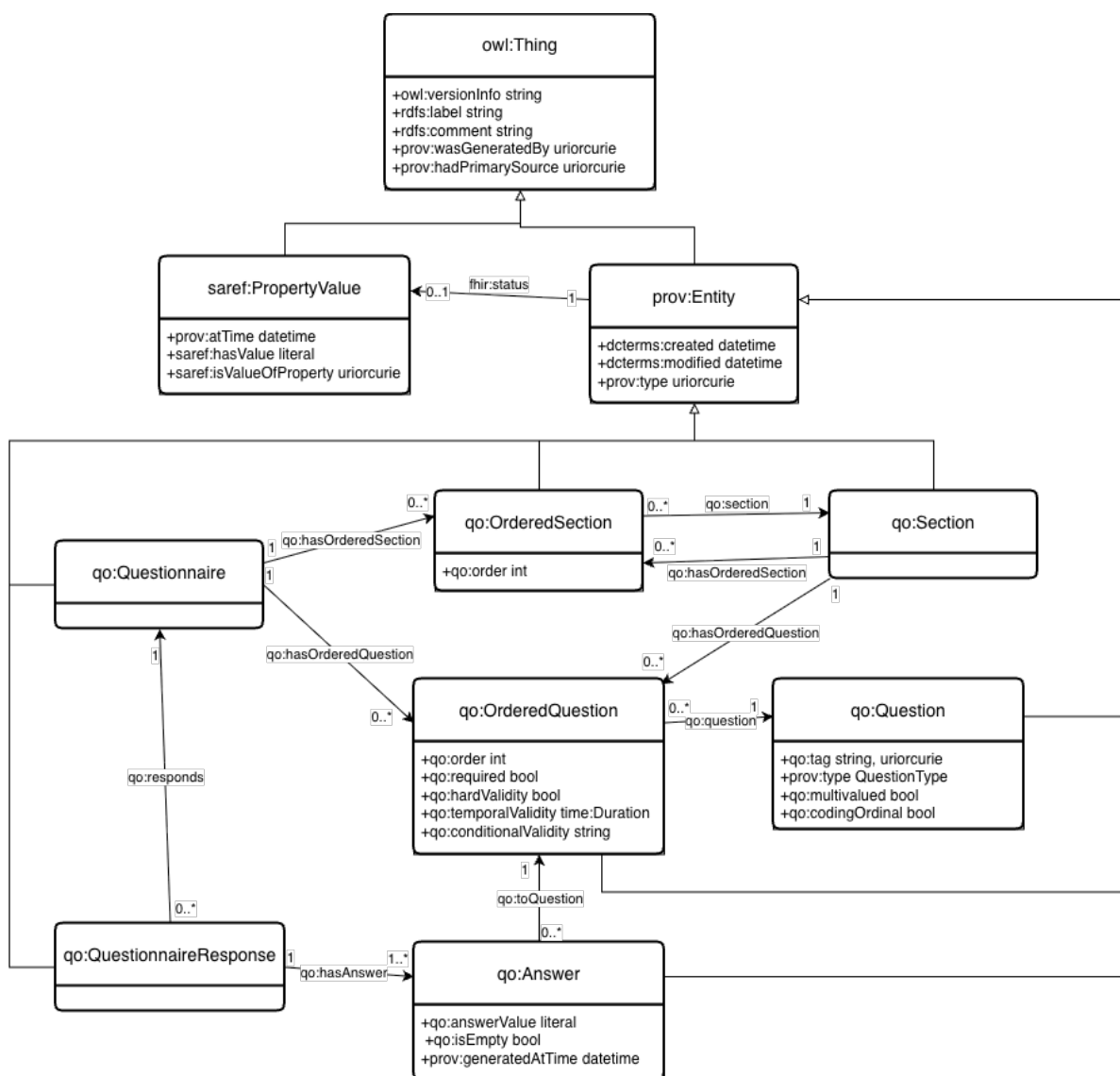


Figure 1: Overview of the FAQIR Questionnaire Ontology architecture, detailing top-level OWL[15], PROV[22] and SAREF[23] grounding, decoupled template structure, and response execution graphs.

²<https://github.com/fairfoundation/questionnaire-ontology>

3.2.1. Upper Ontology Grounding and FAIR Inheritance Rationale

To enforce semantic consistency across all domain entities, the ontology enforces a strict top-level hierarchy rooted in `owl:Thing`. Top-level classes (`foaf:Agent`, `sulo:Process`, `prov:Entity`, and `saref:PropertyValue`) subclass `owl:Thing` directly, guaranteeing that every resource inherits its mandatory global identifiers, human-readable labels (`rdfs:label`), and descriptive comments (`rdfs:comment`), fulfilling the *Findable* and *Accessible* criteria.

Crucially, all survey template entities (`qo:Questionnaire`, `qo:Section`, `qo:Question`, and positional wrappers) as well as execution graphs (`qo:QuestionnaireResponse`, `qo:Answer`) subclass `prov:Entity`. This architectural decision provides immediate semantic interoperability: every questionnaire component inherits optional, standardized slots for attribution (`prov:wasAttributedTo`), creation/modification timestamps (`dcterms:created`, `dcterms:modified`), lifecycle state tracking (`fhir:status`), parthood relationships (`dcterms:hasPart`, `dcterms:isPartOf`), and author attribution (`dcterms:creator`).

To refine these broad inherited properties without breaking parent ontology semantics, QO leverages LinkML's `slot_usage` mechanism. Through `slot_usage`, inherited properties are contextualized per class—enforcing strict presence rules (`required: true`), overriding domain/range expectations (e.g., constraining `dcterms:creator` strictly to a `prov:Organization` range within `qo:Questionnaire`), or enforcing cardinality bounds (`multivalued: false`). This aligns QO directly with W3C PROV-O [22], Dublin Core [24], and ETSI SAREF [23] standards, enabling off-the-shelf Semantic Web reasoners to query provenance, versioning, and structure without requiring proprietary domain rules.

3.2.2. Semantic Vocabulary Reuse and Controlled Value Sets

To maximize FAIR interoperability and eliminate semantic drift, FAQIR systematically reuses established vocabularies, restricting unconstrained literal types (`xsd:string`, numeric scalars) strictly to human documentation labels, contextual rule strings, and raw response payloads (`qo:answerValue`):

- **Question Types:** A question's classification (`prov:type`) aligns with FHIR `ItemType` codes (`choice`, `decimal`, `dateTime`, `text`) [25], extended with custom FAQIR types (`qo:openChoice` and `qo:numberInterval`) to support open selection and bounded numeric inputs.
- **Resource Statuses:** Template and response lifecycle states (`fhir:status`) reuse FHIR status value sets (`draft`, `active`, `retired`, `completed`, `in-progress`) [26].
- **Units of Measure and Temporal Granularity:** Measurement units across numerical parameters (`qo:numericalParams`) and temporal durations (`qo:temporalValidity`) are strictly constrained to standard UCUM (Unified Code for Units of Measure [27]) strings via regex pattern validation (`ucum:`). By importing W3C Time (`time:Duration`) [28], the ontology supports granularities from minute-level acute intervals (`ucum:min`, `ucum:h`) to multi-year chronic validity (`ucum:mo`, `ucum:a`).

3.2.3. Questionnaire and Response Model

The ontology decouples instrument definitions from execution responses, categorizing its core classes into two functional layers:

- **Questionnaire Model:** Structured hierarchically using the root container `qo:Questionnaire`, thematic groupings (`qo:Section`), and context-agnostic inquiry items (`qo:Question`). Questions specify data types via `prov:type` and carry datatype-specific parameter blocks (`qo:codingParams`, `qo:numericalParams`, `qo:intervalParams`). They also record operational flags like `qo:tag` and `qo:multivalued`. Assembling instruments use contextual wrappers (`qo:OrderedSection` and `qo:OrderedQuestion`) that introduce positional sequencing (`qo:order`), required status (`qo:required`), and context-specific validity overrides.

- **Questionnaire Response Model:** Recorded execution data is isolated within `qo:QuestionnaireResponse` (linked to the template via `qo:responds` and attributed to a `foaf:Person` via `prov:wasAttributedTo`). Individual answers (`qo:Answer`) link directly to their originating wrapper (`qo:toQuestion` → `qo:OrderedQuestion`), capturing literal payloads (`qo:answerValue`), explicit omission indicators (`qo:isEmpty`), and execution timestamps (`prov:generatedAtTime`).

3.2.4. Temporal Modeling Rationales

FAQIR incorporates two specific temporal modeling patterns to preserve auditability and reasoning accuracy:

- **Document Creation (`dcterms:created`) vs. Execution Provenance (`prov:generatedAtTime`):** The static administrative timestamp when a template or response session is initialized uses `dcterms:created`. In contrast, individual `qo:Answer` payloads use `prov:generatedAtTime` to record the exact instant a data point was recorded, accommodating asynchronous or multi-session survey completions.
- **saref:PropertyValue for Lifecycle Status Tracking:** Standard direct assertions (e.g., a static `fhir:status` triple) destructively overwrite past resource states upon update. To preserve historical auditability, FAQIR reuses FHIR status value sets alongside SAREF's observational property-value pattern [23]. It introduces custom reified status classes (`qo:QuestionnaireStatus` and `qo:QuestionnaireResponseStatus` subclassing `saref:PropertyValue`) that represent state transitions as explicit graph nodes. Each status node binds the concept URI (e.g., `fhir:active`) via `saref:hasValue`, links to SNOMED CT's Status attribute (`snomed:263490005`) [29] via `saref:isValueOfProperty`, and records the transition timestamp via `prov:atTime`. A status instance is inferred to remain valid until a newer status entity is instantiated, enabling full point-in-time SPARQL auditability.

3.2.5. Formal Constraints, Domains, and Range Axioms

QO graphs are formally constrained at the schema level using LinkML rules compiled into SHACL shapes (`.shacl.ttl`):

- **Domain and Range Bindings:** Object properties strictly enforce domain/range targets to prevent graph dislocation (e.g., $Domain(qo : responds) = qo : QuestionnaireResponse$, $Range(qo : responds) = qo : Questionnaire$, while $Domain(qo : toQuestion) = qo : Answer$ and $Range(qo : toQuestion) = qo : OrderedQuestion$).
- **Type-Driven Conditional Rules:** Preconditions govern parameter requirements based on `prov:type`. Choice-based items ($prov : type \in \{choice, openChoice\}$) require coding blocks ($qo : codingParams \geq 1$), while interval items ($prov : type = numberInterval$) mandate both $qo : numericalParams$ and $qo : intervalParams$.
- **Sequence Uniqueness:** Contextual wrappers (`qo:OrderedQuestion`, `qo:OrderedSection`) enforce explicit uniqueness constraints on sequence positioning (`qo:order`) within any container instance.

3.3. Dual-Level Modeling Rationale and Blank Node Mitigation

FAQIR follows a **dual-level modeling** approach separating content definition from questionnaire structure. Atomic questions (`qo:Question`) and sections (`qo:Section`) are defined once as reusable entities and assembled into instruments using ordered wrappers (`qo:OrderedQuestion`, `qo:OrderedSection`). This mirrors standard archetype patterns (e.g., openEHR [30]) while preventing structural fragmentation.

This decoupled design requires custom classes rather than direct FHIR RDF re-use. FHIR is designed for client-server data exchange rather than graph-based longitudinal reasoning; when expressed in

RDF, FHIR relies heavily on *blank nodes* for nested structures, hindering global identification, graph discovery, and direct SPARQL traversal over historical answers [12]. FAQIR assigns global URIs to all template containers, questions, contextual wrappers, and response instances, eliminating blank nodes across the structural graph. Anonymous blank nodes are restricted strictly to dependent, inline datatype parameters (such as `fhir:Coding` selection blocks) that do not require independent global referencing.

Linking `qo:Answer` to `qo:OrderedQuestion` (the wrapper) rather than directly to `qo:Question` preserves three key distinctions: (1) disambiguating atomic questions that appear multiple times in a single survey (e.g., pre- and post-intervention pain), (2) attaching context-specific validity rules (e.g., 6-month validity in routine screening vs. 1-week validity pre-surgery), and (3) properties like required status and order position belong to the act of asking, not to the question itself. Global question reusability is maintained via graph traversal: `Answer` $\rightarrow$ `OrderedQuestion` $\rightarrow$ `Question`.

3.4. Temporal Answer Validity and Reasoning

Temporal validity defines the timeframe during which a recorded answer remains *current*, *trustworthy*, and *actionable*. Validity depends on both the atomic question semantics and the administration context. To support this, validity is defined across five hierarchical levels where narrower contexts override broader ones: `qo:Question` (broadest) $\rightarrow$ `qo:OrderedQuestion` $\rightarrow$ `qo:Section` $\rightarrow$ `qo:OrderedSection` $\rightarrow$ `qo:Questionnaire` (most restrictive).

Each level carries three machine-actionable validity attributes:

- `qo:temporalValidity` (`time:Duration`): The machine-actionable temporal lifespan of an answer.
- `qo:hardValidity` (Boolean): Specifies operational enforcement upon expiration:
 - **Strict Deadline** (`true`): Bounded by explicit phrase semantics or clinical constraints (e.g., questions asking about "TODAY" carry a maximum 1-day P1D hard validity). Once expired, the answer is strictly invalidated and *must* be re-asked.
 - **Orientative Guideline** (`false`): Bounded by soft heuristics (e.g., general satisfaction items carrying an estimated 1-year P1Y soft validity). Upon expiration, the prior response is pre-filled, prompting the patient only for confirmation.
- `qo:conditionalValidity` (String): Machine-readable rule conditions defining early revocation events (e.g., a documented smoking cessation intervention invalidating a prior response to "Do you smoke?").

During evaluation, the reasoning engine compares an answer's execution timestamp (`prov:generatedAtTime`) against the evaluation timestamp plus its context-specific `qo:temporalValidity`. If within the window and unrevoked by `qo:conditionalValidity`, the answer is active. If expired, the engine evaluates `qo:hardValidity`: forcing mandatory re-administration if `true`, or flagging for automated pre-filling if `false`.

3.5. Consent-Enforced Deferral and Fatigue Reduction

To ensure patient autonomy, the query and reasoning engine enforces consent checking using ODRL (Open Digital Rights Language) and DPV (Data Privacy Vocabulary) [31, 32]. When a patient initiates a questionnaire, the system evaluates active READ permissions for past responses. If consent is absent, questions are re-administered regardless of temporal validity.

When past answers are both temporally valid and covered by active READ consent, the system automatically defers redundant entries by either (1) automatically answering the item without presenting it, or (2) pre-filling the response for rapid confirmation (`qo:hardValidity` = `false`). This directly reduces questionnaire completion time and mitigates survey fatigue while upholding strict data governance.

4. Clinical Validation: WHOQOL-BREF Use Case

To evaluate the expressiveness, semantic integrity, and operational feasibility of the FAQIR Questionnaire Ontology (QO), we instantiated two real-world, validated Patient-Reported Outcome Measures (PROMs): the World Health Organization Quality of Life instrument (WHOQOL-BREF) [8] and the EuroQol EQ-5D-5L [7].

The WHOQOL-BREF comprises 26 items categorized across four quality-of-life domains (Physical, Psychological, Social Relationships, and Environment). Demonstrating QO on WHOQOL-BREF proves the model's capacity to handle complex, multi-domain instruments without data duplication or structural fragmentation.

4.1. Instrument Modeling and Graph Instantiation

To operationalize the WHOQOL-BREF instrument, we developed an automated Python transformation script that parses language-specific PDF questionnaires (e.g., English and Spanish versions) and generates canonical RDF Turtle instantiations aligned with the FAQIR ontology. This automated pipeline extracts localized question labels and response scales, binding multilingual RDF literals (@en, @es) directly to context-agnostic `qo:Question` entities without duplicating underlying concept identifiers.

Following the dual-level architecture described in Section 3, the 26 items of the WHOQOL-BREF were instantiated as 26 context-agnostic `qo:Question` entities in RDF Turtle format. Each question is assigned a global URI under the FAQIR namespace, annotated with multilingual labels (`rdfs:label`), tagged with standardized mnemonic codes (`qo:tag`), and parameterized using FHIR ItemType codes (`prov:type = choice`) mapped to 5-point Likert scale[33] options (`qo:codingParams`).

The WHOQOL-BREF survey template (`qo:Questionnaire`) aggregates four thematic sections (`qo:Section`), which in turn hold 26 contextual wrappers (`qo:OrderedQuestion`). Positional indices (`qo:order`) and default temporal lifespans (`qo:temporalValidity = P2W`, representing a two-week validity window common for acute quality-of-life assessments) are attached strictly to the wrappers.

Listing 1 illustrates a representative RDF snippet showing the separation between the atomic question entity, its contextual wrapper, and an executed patient response.

Listing 1: Representative RDF instantiation of the WHOQOL-BREF demonstrating decoupled atomic questions, contextual wrappers, and patient response payloads.

```
# Atomic Context-Agnostic Question Definition
faqir:question_WHOQOL_BREF_01 a qo:Question ;
  rdfs:label "How would you rate your quality of life?"@en ;
  qo:tag snomed:405152002 ; # Quality of life satisfaction (observable entity)
  prov:type <http://hl7.org/fhir/item-type#coding> ;
  qo:codingParams [ a fhir:Coding ; fhir:code "1" ; fhir:display "Very poor"@en ],
    #...
    [ a fhir:Coding ; fhir:code "5" ; fhir:display "Very good"@en ] ;
  dcterms:creator <https://www.who.int> .

# Contextual Ordered Wrapper within WHOQOL-BREF Template
faqir:orderedquestion_WHOQOL_BREF_01 a qo:OrderedQuestion ;
  qo:question faqir:question_WHOQOL_BREF_01 ;
  qo:order 1 ;
  qo:required true ;
  qo:hardValidity false ;
  qo:temporalValidity faqir:duration_14days .

faqir:duration_14days a time:Duration ;
  saref:hasValue 14.0 ;
  saref:isMeasuredIn ucum:d .

# Execution Response Instance
faqir:response_WHOQOL_BREF_01 a qo:QuestionnaireResponse ;
  prov:wasAttributedTo <https://pods.faqir.org/234> ;
  qo:hasAnswer faqir:answer_WHOQOL_BREF_01_a01 .

faqir:answer_WHOQOL_BREF_01_a01 a qo:Answer ;
  qo:toQuestion faqir:orderedquestion_WHOQOL_BREF_01 ;
```

```

qo:isEmpty false ;
prov:generatedAtTime "2026-06-30T14:57:33.168913+00:00"^^xsd:dateTime ;
qo:answerValue "1" .

```

4.2. Longitudinal Reasoning and Fatigue Reduction Walk-Through

To validate the graph architecture and demonstrate question deferral in clinical practice, we executed a synthetic longitudinal patient scenario across three sequential clinical encounters over a two-month period. Note that because full automated ODRL-DPV policy reasoning is currently under active development (and framed as future work in Section 6), graph queries at present evaluate temporal answer validity under the assumption that active `odr1:Read` consent has been granted:

1. **Initial Visit ($T_0 = \text{Day 0}$):** Patient 123 completes the baseline WHOQOL-BREF instrument. The KG construction module instantiates a `qo:QuestionnaireResponse` containing 26 individual `qo:Answer` nodes, each capturing its exact execution timestamp (`prov:generatedAtTime = T_0`).
2. **Follow-Up Visit 1 ($T_1 = \text{Day 10}$):** The patient returns for a follow-up appointment 10 days later. Upon initiating a new survey request, the reasoning service executes temporal queries corresponding to CQ3 and CQ4. It evaluates $T_1 - T_0 = 10$ days, which falls strictly within the 14-day validity lifespan (P2W). Assuming active read permissions, all 26 answers remain valid. For items with soft validity (`qo:hardValidity = false`), the system pre-fills prior answers and presents a single summary screen for rapid patient confirmation. This reduces active typing burden by 100% while ensuring clinical accuracy.
3. **Follow-Up Visit 2 ($T_2 = \text{Day 45}$):** The patient returns 45 days after baseline. The reasoning engine evaluates $T_2 - T_0 = 45$ days $>$ P2W. The temporal validity window has elapsed. Consequently, past answers are invalidated, flagging all 26 questions for mandatory re-administration.

4.3. Verification of Competency Questions and Artifacts

The instantiated WHOQOL-BREF and EQ-5D-5L knowledge graphs were verified against all Competency Questions (CQ1–CQ5) using executable SPARQL queries hosted in the project repository. Query execution confirmed that:

- **CQ1 & CQ2 (Composition and Constraints):** The engine successfully reconstructs full survey layouts in exact numerical sequence (`qo:order`) while retrieving type-specific range constraints.
- **CQ3 & CQ4 (Validity and Deferral):** The engine accurately identifies active vs. expired answer sets based on execution timestamps and filters deferrable questions under granted read consent assumptions.
- **CQ5 (Cross-Instrument Analytics):** SPARQL queries successfully isolate atomic questions shared across different instruments (e.g., matching mobility items across EQ-5D-5L and WHOQOL-BREF via matching `qo:tag` identifiers and SNOMED CT concepts).

All compiled OWL ontologies, SHACL validation shapes, synthetic test graphs, and executable SPARQL query files are openly available for review and execution in the project repository.

5. Discussion and Related Work

This paper makes four primary ontological and technical contributions:

1. **Decoupled, FAIR-Compliant Architecture:** A dual-level, FAIR-by-design ontology (QO) that decouples atomic inquiry definitions from contextual questionnaire layout, eliminating blank nodes and transforming static survey documents into queryable, machine-readable knowledge graphs.

2. **Machine-Actionable Temporal Validity Reasoning:** The first semantic framework to operationalize context-sensitive answer lifetimes (`qo:temporalValidity`) and enforcement mechanisms (`qo:hardValidity`, `qo:conditionalValidity`) to calculate answer trustworthiness over time.
3. **Consent-Enforced Fatigue Reduction:** A patient-centric workflow combining temporal validity checking with fine-grained ODRL-DPV[32, 31] consent enforcement to automatically defer or pre-fill valid historical answers, directly mitigating question-survey fatigue while safeguarding patient autonomy.
4. **Reusable Question Catalogue & Cross-Walk Support:** By treating atomic questions (`qo:Question`) as independent, context-agnostic entities carrying machine-readable tags (`qo:tag`) mapped to controlled clinical vocabularies (e.g., SNOMED CT [20]), QO enables the creation of a centralized, reusable question catalogue that spans across validated instruments, facilitating standardized data capture and automated cross-walk analysis.

To ensure broad community findability, accessibility, and reuse, the ontology has been engineered according to FAIR principles and is prepared for submission to BioPortal³. Crucially, while FAQIR is validated within the healthcare domain using Patient-Reported Outcome Measures (PROMs) such as EQ-5D-5L and WHOQOL-BREF, the core Question-Ontology (QO) architecture is entirely domain-agnostic. The separation between atomic questions, contextual wrappers, and temporal validity reasoning can be directly applied to any structured questionnaire application, including sociological surveys, educational assessments, and customer feedback systems.

5.1. Comparison with State-of-the-Art Ontologies

While several ontologies address survey representation, FAQIR deliberately refrains from inheriting directly from existing models (such as QUESTO or QuestionOnto) because aligning with their core assumptions would introduce fundamental architectural limitations.

First, **QUESTO** [13] models survey items under the Information Artifact Ontology (IAO) hierarchy, blurring the distinction between an inquiry specification (`Directive Information Entity`) and its recorded response (`Information Content Entity`). Crucially, like **QuestionOnto** [14], QUESTO conflates self-reported survey answers with objective clinical observations within the same structural representation. FAQIR enforces a strict conceptual separation: a `qo:Question` defines an information request, a `qo:Answer` records a self-reported reply, and objective physical measurements (e.g., blood pressure, lab results) must be captured as formal clinical observations (e.g., via SAREF) rather than survey items. Aligning either ontology to this model would require breaking changes to their class hierarchies and property domains.

Second, beyond conflating observations and questions, **QuestionOnto** [14] embeds contextual metadata—such as positional sequence indices and spatial layout instructions—directly within atomic question instances and answer labels. This design causes massive data duplication across survey administrations and severely hinders cross-context reasoning. By contrast, FAQIR offloads all contextual metadata to lightweight wrapper classes (`qo:OrderedQuestion` and `qo:OrderedSection`). An atomic `qo:Question` remains context-agnostic and globally reusable across instruments, while positional ordering (`qo:order`), required status (`qo:required`), and temporal validity thresholds belong strictly to the wrapper. Adapting QuestionOnto to this decoupled structure would break its existing instance schemas and query interfaces.

To our knowledge, FAQIR is the first operationalized ontology that combines dual-level survey modeling with machine-actionable temporal validity reasoning and explicit, consent-enforced execution rules.

³<https://biportal.bioontology.org/>

5.2. Limitations and Trade-offs

Despite its strengths, FAQIR acknowledges several technical and operational trade-offs: **Abstraction of Temporal Validity:** Determining the exact validity duration of an answer is inherently heuristic [34]. While clinical consensus may suggest that an answer to "Do you smoke?" remains valid for six months, validity periods vary by institutional guidelines and patient context. The `qo:hardValidity` flag cleanly distinguishes strict expiration deadlines (e.g., EQ-5D-5L items assessing health "TODAY") from orientative guidelines (e.g., lifestyle satisfaction), but both require domain expert configuration. FAQIR provides the machine-actionable framework to represent and reason over these rules, but it does not prescribe the clinical guidelines themselves.

Usability vs. Privacy in Consent Management: Enforcing ODRL-DPV READ permissions before reusing historical answers is essential for privacy compliance. However, requiring granular permissions for every question could overwhelm users, causing "consent fatigue" or driving blanket approvals that undermine meaningful choice. Future work will investigate user-friendly consent mechanisms, such as category-level permissions (e.g., "allow reuse for all lifestyle questions") and dynamic privacy dashboards, alongside expanding the reasoning engine to support full ODRL policy evaluation.

6. Conclusions and Future Work

6.1. Conclusions

This paper presented the FAQIR Questionnaire Ontology (QO), a FAIR-by-design framework that transforms static, document-based survey instruments into machine-readable, queryable knowledge graphs. By enforcing a dual-level architecture that decouples context-agnostic inquiry definitions (`qo:Question`) from positional survey layout (`qo:OrderedQuestion`), QO eliminates blank nodes across core structural entities, prevents data duplication, and enables global answer reusability. Grounding top-level classes in W3C PROV-O, Dublin Core, and ETSI SAREF provides out-of-the-box interoperability and preserves full historical auditability through reified status tracking (`saref:PropertyValue`).

Crucially, FAQIR introduces machine-actionable temporal validity reasoning (`qo:temporalValidity`, `qo:hardValidity`) combined with fine-grained ODRL-DPV consent enforcement. This enables active, patient-centric survey management that automatically defers or pre-fills valid historical answers. FAQIR redefines questionnaires as living, consent-aware knowledge graphs, offering a concrete, scalable path to reduce repetitive survey fatigue while improving longitudinal data quality in healthcare.

6.2. Future Work

Building upon the fully implemented Knowledge Graph construction module, future development will progress across five key directions:

1. **Reasoning Engine Completion and Clinical Validation:** Completing and optimizing the SPARQL-based temporal reasoning engine to evaluate context-specific lifespan rules and execute ODRL-DPV permission checks in real time. Following deployment, QO will be evaluated on real-world longitudinal patient cohorts to measure actual question burden reduction, validate the accuracy of validity heuristics, and assess patient acceptance.
2. **Ontology Governance, Maintenance, and BioPortal Community Release:** Sustaining the long-term usability of QO requires continuous ontology governance. Essential future work includes maintaining the canonical LinkML source schema, updating auto-generated SHACL shapes and documentation artifacts upon schema evolution, and finalizing submission to BioPortal to foster open community adoption and alignment with emerging healthcare data standards.
3. **Patient-Centric Consent Usability:** Co-designing privacy management interfaces with patients and clinical stakeholders. Future iterations will explore tiered, category-level permission controls

(e.g., granting blanket approval for routine lifestyle items) and dynamic consent dashboards to prevent "consent fatigue" while safeguarding patient autonomy.

4. **LLM-Driven Phrasing Diversification:** Leveraging Large Language Models (LLMs) to generate linguistically diverse question phrasings without altering underlying semantics [35]. FAQIR's separation of atomic `qo:Question` nodes from presentation wrappers (`qo:OrderedQuestion`) provides a natural foundation: multiple phrasing variants (e.g., "Do you smoke?" vs. "Are you currently a tobacco user?") can link to a single atomic question entity. This will allow dynamic, LLM-driven presentation to reduce perceived survey monotony while preserving strict psychometric and answer reusability.
5. **Ecosystem Expansion via a Public Question Catalogue:** Expanding the use of `qo:tag` fields mapped to international clinical terminologies (e.g., SNOMED CT [20]) to establish a public, cross-institutional question catalogue. This will allow researchers and clinicians to discover, share, and cross-walk standardized survey items across validated instruments, transitioning FAQIR from a standalone ontology into a collaborative semantic ecosystem.

Declaration on Generative AI

During the preparation of this work, the authors used Gemini and ChatGPT in order to: Grammar and spelling check. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

Ethics and Environmental Footprint

This work uses only synthetic patient data; no ethics approval was required. Patient consent is enforced via ODRL-DPV permissions before any historical answer is accessed. The lightweight ontology and reasoning engine have minimal expected computational overhead.

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